

Rezwan N. Hoque

CONTACT INFORMATION	https://rezrezrezrez.github.io/	rezwan.hoque98@bcmail.cuny.edu
EDUCATION	Brooklyn College (CUNY) , Brooklyn, NY B.S. in Mathematics (Theoretical Concentration) GPA: 3.983	Expected May 2026
HONORS & AWARDS	PME Travel Grant (Fall 2025) Zimmerberg, Hyman J. '41 Fund for Undergraduate Research in Mathematics (Spring 2024, Spring 2025) Tow Mentoring Initiative Mentee (Spring 2024) Dean's List (since Fall 2022)	
PAPERS & PUBLICATIONS	Degeneracy and Sato-Tate Groups of $y^2 = x^{p^2} - 1$. With Justin Chen, Heidi Goodson, and Sabeeha Malikhah. Accepted for publication 2026. arXiv: https://arxiv.org/abs/2512.03299v1 Sato-Tate Groups and Distributions of $y^\ell = x(x^\ell - 1)$. With Heidi Goodson. <i>Acta Arithmetica</i> . 221(4): 355-370 (2025)	
PRESENTATIONS	Sato-Tate Groups and Distributions of $y^2 = x^{p^2} - 1$ PME Contributed Session on Research by Undergraduates, Joint Mathematics Meetings (January 2026) Sato-Tate Groups and Distributions of $y^\ell = x(x^\ell - 1)$ AMS-PME Poster Session, Joint Mathematics Meetings (January 2025) Sato-Tate Groups of Aoki Curves Tow Mentoring and Research Student Conference (May 2024) Decimal Expansions: How can we Skip Long Division? CUNY DRP Poster Day (May 2023)	
RESEARCH	Brooklyn College (CUNY) Brooklyn, NY Research Assistant • Collaborated with Prof. Heidi Goodson, PhD • Determined Sato-Tate groups and distributions of nondegenerate curves $y^\ell = x(x^\ell - 1)$ and degenerate curves $y^2 = x^{p^2} - 1$ using algebraic and arithmetic geometry • Computed theoretical moment statistics of normalized L-polynomial coefficients via Sage and Python, optimized for $\sim 97\%$ reduction in runtime	January 2024 - January 2026
SERVICE	Member , Brooklyn College Department of Mathematics Curriculum Committee (March 2024 - Present) Secretary , Mathematics Club of Brooklyn College (January 2024 - May 2025)	

EMPLOYMENT
& EXPERIENCE

CRC/SBS Summer Institute REU

Brooklyn, NY

Mentee

July 2025 - August 2025

- Worked under Dr. Terrence R. Blackman, Chair of Mathematics Department at Medgar Evers College
- Performed editorial work on a forthcoming biographical anthology presenting six contemporary African American mathematicians and their technical works

CUNY Directed Reading Program

New York, NY

Mentee

January 2023 - May 2023

- Studied elementary and algebraic number theory
- Presented findings on retrieving blocks of non-terminating decimal expansions and necessary and sufficient conditions for terminating decimal expansions

Mathnasium

Brooklyn, NY

Tutor

December 2021 - June 2022

- Mentored K-12 students in math via personalized plans